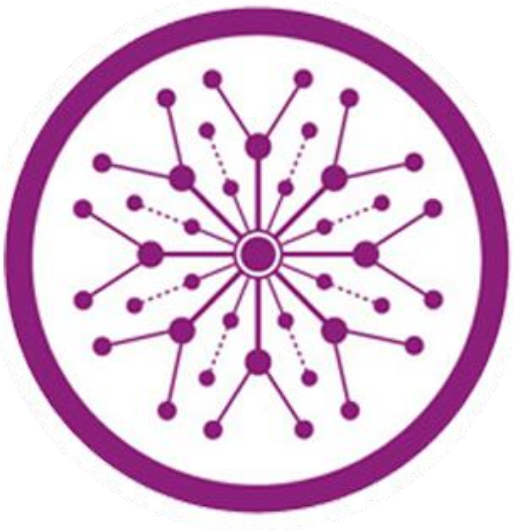
ReqTrace AI

Final Year Project

Session 2023-2027

A project submitted in partial fulfillment of the degree of

BS in Software Engineering



Department of Software Engineering

Faculty of Computer Science & Information Technology

The Superior University, Lahore

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*The candidates confirm that the work submitted is their own and appropriate credit has been given where reference has been made to work of others

Plagiarism Free Certificate

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Project Report

ReqTrace AI

Change Record

Author(s)	Version	Date	Notes	Supervisor's Signature
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Kamran	1.1	13-10-2026	Initial Documentation Setup	
Areeba	1.2	14-10-2026	Document Revisions & Chapter 1 Updates	

APPROVAL

PROJECT SUPERVISOR

Comments: _____

_____

Name: _____

Date: _____ Signature: _____

PROJECT MANAGER

Comments: _____

_____

Name: _____

Date: _____ Signature: _____

HEAD OF DEPARTMENT

Comments: _____

_____

Name: _____

Date: _____ Signature: _____

Dedication

This work is dedicated to our beloved parents, whose unconditional love, endless support, and encouragement have been our greatest source of strength throughout our academic journey. We also dedicate this project to our respected teachers, who have inspired us to pursue knowledge, overcome challenges, and strive for excellence. Their guidance and motivation have played a significant role in our achievements.

Acknowledgements

First and foremost, we express our sincere gratitude to Almighty Allah for granting us the strength, wisdom, and determination to complete our Final Year Project, ReqTrace AI. We are deeply thankful to our respected supervisor for their continuous guidance, valuable suggestions, constructive feedback, and encouragement throughout this project. Their expertise and support have played a significant role in shaping our ideas and improving the quality of our work. We would also like to extend our appreciation to the faculty members of the Department of Software Engineering at The Superior University, Lahore, for providing us with the essential knowledge and skills required for this project. Finally, we are sincerely grateful to our parents, families, and friends for their unconditional love, patience, motivation, and constant support throughout our academic journey.

Executive Summary

ReqTrace AI is an Artificial Intelligence (AI)-powered web-based platform designed to improve software requirements management and minimize scope creep in software development projects. Software houses often receive client requirements through informal communication channels such as WhatsApp, emails, and text conversations, which can lead to undocumented changes, misunderstandings, project delays, and disputes. ReqTrace AI addresses these challenges by converting unstructured client communication into organized, traceable, and approved software requirements.

The system utilizes Natural Language Processing (NLP), Large Language Models (LLMs), and Retrieval-Augmented Generation (RAG) to extract atomic requirements from client conversations in English, Roman Urdu, and mixed languages. Each extracted requirement is linked to its original source message, ensuring transparency and accuracy. A human-in-the-loop review process allows project managers to approve, edit, or reject AI-generated requirements. The platform maintains version histories and immutable requirement baselines, enabling intelligent scope-change detection by comparing new client requests against previously approved requirements.

ReqTrace AI also incorporates a tamper-evident audit trail using cryptographic hash chaining to maintain reliable records of requirement changes and approvals. Additional features include client sign-off, Jira integration for requirement-to-task traceability, role-based access control, PII masking, change-request message generation, and PDF handover reports. The system is developed using modern technologies, including Next.js, TypeScript, Python FastAPI, PostgreSQL, LangChain, and AI models supported through GroqCloud and Ollama.

The proposed system will be evaluated through requirement extraction accuracy, classification F1-score, scope-change detection performance, data privacy effectiveness, audit integrity, and usability testing. Real-world project datasets and pilot testing with software development teams will be used to assess its effectiveness. Ultimately, ReqTrace AI aims to reduce undocumented scope changes, improve requirements traceability, strengthen client-developer communication, and provide a reliable and efficient requirements management solution for software houses and freelance development teams.

List of Contents

Dedication v
Acknowledgements vi
Executive Summary vii
List of Figures ix
List of Tables x
Chapter 1 1
Introduction 1
1.1. Background 2
1.2. Motivations and Challenges 2
1.3. Goals and Objectives 2
1.4. Literature Review/Existing Solutions 3
1.6. Proposed Solution 4
1.7. Project Plan 4
1.7.1. Work Breakdown Structure 5
1.7.2. Roles & Responsibility Matrix 6
1.7.3. Gantt Chart 8
1.8. Report Outline 9
1.9. Empathy Map 10

List of Figures

Figure 1. Gantt Chart.....9

List of Tables

Table 1. Gap Analysis of Existing Solutions 3
Table 2. Work Breakdown Structure of ReqTrace AI 5
Table 3. Roles & Responsibility Matrix 7
Table 4. Empathy Map of the Project Manager 10

Chapter 1
Introduction

Chapter 1: Introduction

ReqTrace AI is an AI-powered web-based platform designed to improve software requirements management and control scope creep. It converts informal client communication from WhatsApp, emails, and text conversations into structured, traceable requirements. The system helps software development teams manage requirement changes, maintain transparency, and reduce misunderstandings between clients and developers.

1.1.Background

In software development, requirements are essential for successful project delivery. However, many software houses manage client requirements through informal conversations, making documentation and change tracking difficult. This often results in unclear requirements, undocumented changes, and project delays. ReqTrace AI addresses these issues through AI-assisted requirement extraction, approval workflows, and scope monitoring.

1.2.Motivations and Challenges

The main motivation behind ReqTrace AI is to reduce manual requirement documentation and prevent uncontrolled scope changes. Software teams frequently struggle with scattered conversations, ambiguous client requests, and missing approval records. Processing English and Roman Urdu conversations, protecting sensitive data, and accurately identifying requirements are additional challenges. The proposed system aims to address these problems through AI assistance, human verification, and secure requirement tracking.

1.3.Goals and Objectives

The main goal of ReqTrace AI is to develop an intelligent and reliable requirements management platform.

The key objectives are:

1. Extract structured requirements from WhatsApp chats, emails, and text conversations.

2. Maintain requirement traceability, version history, and approved baselines.
3. Detect and classify scope changes using AI.
4. Maintain tamper-evident audit records and client approvals.
5. Integrate Jira for requirement-to-task traceability.
6. Ensure data privacy, human verification, and measurable system accuracy.

1.4.Literature Review/Existing Solutions

Existing tools such as Jira and Asana support project and task management, while Notion and Confluence provide documentation and collaboration features. Traditional requirements management systems offer structured requirement tracking and change management. AI-based summarization tools can analyze conversations and generate summaries. However, these solutions do not necessarily combine informal chat-based requirement extraction, human approval, automated scope-change detection, and tamper-evident evidence management in a single workflow designed for small software teams.

1.5.Gap Analysis

Existing solutions address different aspects of project management but often require additional tools or manual effort for complete requirements management.

Table 1 *Gap Analysis of Existing Solutions*

Feature	Jira / Asana	Notion / Confluence	ReqTrace AI (Proposed)
Chat-based requirement extraction	Not built-in	Mostly manual	AI-powered
Requirement traceability	Supported through issue links	Manual links	Automatic source links
Scope-change detection	Requires configuration	Not built-in	AI-based classification
Immutable baselines	Requires additional controls	Not built-in	Supported

Tamper-evident audit trail	Activity history	Version history	Hash-chain verification
Client sign-off	Requires configuration	External workflow	Integrated approval

ReqTrace AI aims to bridge these gaps by combining AI-assisted requirement extraction, traceability, scope-change detection, and verifiable approvals in one centralized platform.

1.6. Proposed Solution

ReqTrace AI proposes an AI-powered web application that extracts requirements from client conversations using Natural Language Processing (NLP) and Large Language Models (LLMs). Project managers can review, approve, edit, or reject extracted requirements. The system maintains versioned SRS documents, immutable baselines, and AI-based scope-change alerts. It also includes secure data handling, a tamper-evident audit trail, client sign-off, Jira integration, and handover reports.

The platform will be developed using Next.js, TypeScript, Python FastAPI, PostgreSQL, LangChain, GroqCloud, and Ollama. Its performance will be evaluated through AI accuracy, security, traceability, and usability testing.

1.7.Project Plan

The development of ReqTrace AI is planned over a period of 12 months, following a phased and iterative software development approach. The project focuses on designing and implementing an AI-powered requirements management platform that automates requirement extraction, maintains traceability, detects scope changes, and provides verifiable approval records.

The development process is divided into research and planning, system design, frontend and backend development, AI integration, security implementation, testing, and deployment. Responsibilities are distributed among three team members to ensure balanced participation and effective collaboration. The project will prioritize core engineering features and their evaluation before introducing additional functionalities.

1.7.1. Work Breakdown Structure

The Work Breakdown Structure (WBS) provides a hierarchical representation of the major deliverables and activities required to complete ReqTrace AI. It divides the project into eight major work packages, each containing specific tasks. This structure supports task assignment, progress monitoring, and effective resource management throughout the development lifecycle.

Table 2 *Work Breakdown Structure of ReqTrace AI*

WBS #	WBS Deliverable	Activity #	Activity Description	Duration (Days)	Responsible Member(s)
1	Project Management	1.1	Project planning and WBS preparation	3	Areeba
1	Project Management	1.2	Requirements analysis and stakeholder interviews	12	Areeba
1	Project Management	1.3	Literature review and documentation	7	Areeba, Kamran
2	System Design	2.1	System architecture and database design	8	Usman
2	System Design	2.2	UI/UX wireframes and interface design	10	Areeba
3	Dataset Preparation	3.1	Data collection and anonymization	20	Areeba, Kamran
3	Dataset Preparation	3.2	Data annotation and validation	25	Kamran, Areeba
4	Application Development	4.1	Frontend development and review queue	25	Areeba
4	Application Development	4.2	Backend APIs, authentication and database	25	Usman

4	Application Development	4.3	Chat ingestion and PII masking	12	Usman, Kamran
5	AI Development	5.1	AI requirement extraction and provenance	20	Kamran
5	AI Development	5.2	Requirement matching, versioning and baselines	18	Usman, Kamran
5	AI Development	5.3	Scope-change classification using RAG	18	Kamran
6	Security & Integration	6.1	Audit trail and client sign-off	15	Usman
6	Security & Integration	6.2	Gmail and Jira integration	12	Usman, Kamran
6	Security & Integration	6.3	Change-request generator and handover PDF	8	Areeba, Kamran
7	Testing & Deployment	7.1	Functional, AI, security and performance testing	20	All Members
7	Testing & Deployment	7.2	Pilot deployment and usability evaluation	20	All Members
7	Testing & Deployment	7.3	Final documentation and presentation	12	All Members

Note: Durations are estimated working days. Several activities will run in parallel within the 12-month project timeline. The 20-day pilot activity represents approximately four working weeks.

1.7.2. Roles & Responsibility Matrix

1. Project Lead & Backend Developer (Areeba):

- Manages project planning, task allocation, and overall development progress.

- Leads backend development using Python FastAPI, database management, and API integration.
- Oversees AI integration, system architecture, security, and core functionality.

2. Frontend Developer (Usman):

- Responsible for UI/UX design and frontend development using Next.js, TypeScript, and Tailwind CSS.
- Develops dashboards, review queues, requirement management screens, and other user interfaces.
- Ensures responsive design and seamless frontend-backend integration.

3. Quality Assurance & Documentation Lead (Kamran):

- Performs functional, integration, security, and AI performance testing.
- Ensures the system meets functional and non-functional requirements.
- Prepares technical documentation, test reports, and evaluation results.

4. Project Supervisor (Mr. Hafiz Ali Ahmad):

- Provides academic guidance and technical recommendations throughout the project.
- Reviews project progress, documentation, and implementation quality.
- Ensures the project meets academic standards and defined objectives.

Table 3 *Roles & Responsibility Matrix*

Project Activity	Areeba	Usman	Kamran
Project Planning & Management	P	S	S
Backend & Database Development	P	S	—
AI Integration & Core Features	P	S	S
Frontend & UI/UX Development	S	P	—
API Integration & Security	P	S	S

Software Testing & Quality Assurance	S	S	P
AI Evaluation & Performance Testing	S	S	P
Documentation & Reporting	S	S	P
Final Integration & Presentation	P	S	S

P = Primary Responsibility,

S = Supporting Responsibility,

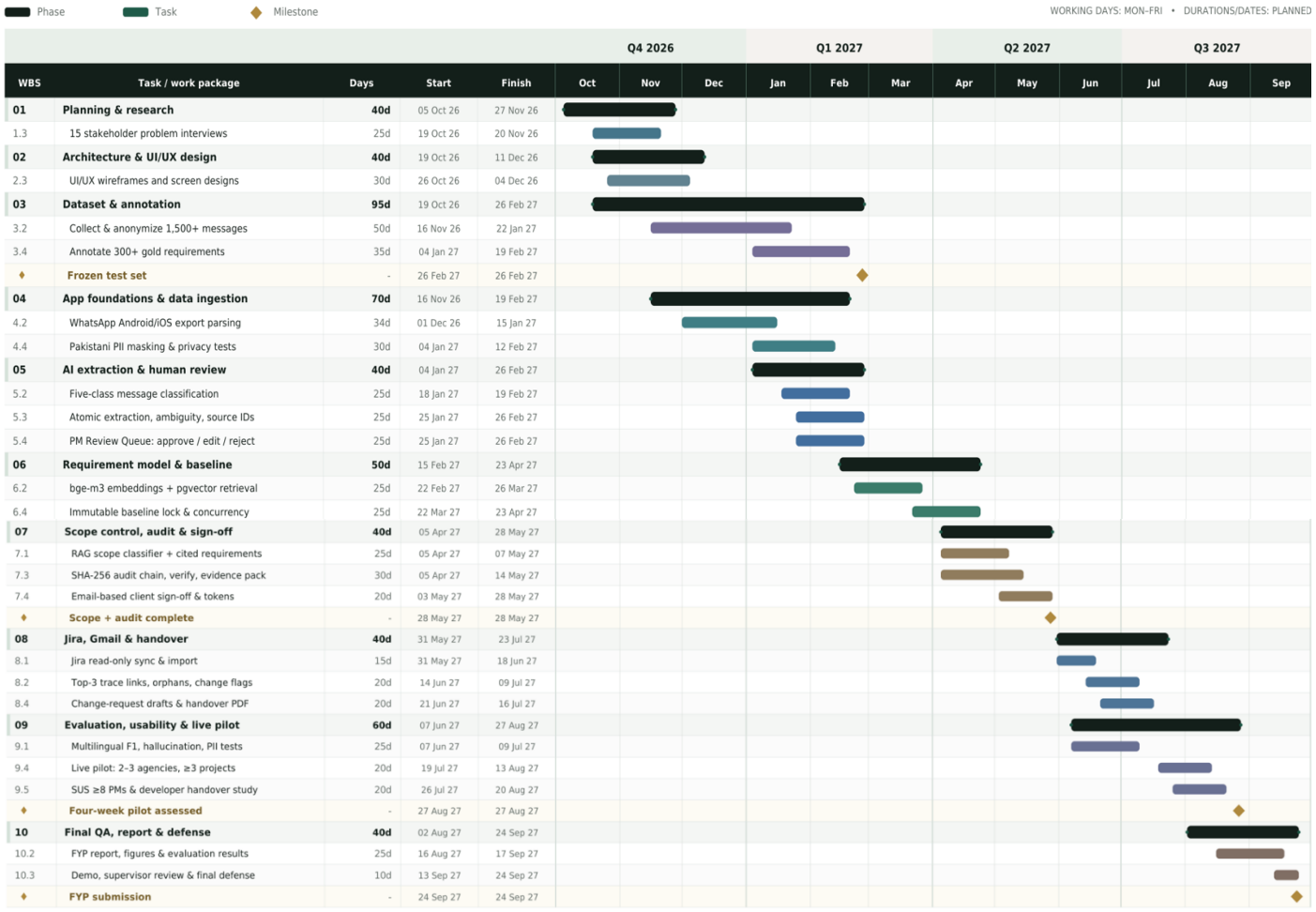
— = Not Directly Assigned.

The **project supervisor** will provide guidance and review progress throughout the development process. All three team members will collaborate to ensure the successful completion of ReqTrace AI.

1.7.3. Gantt Chart

The Gantt Chart illustrates the scheduled activities of ReqTrace AI over the 12-month development period. It helps track project progress, task durations, and overlapping development phases.

Figure 1 *Gantt Chart*



WBS	Task / work package	Days	Start	Finish
01	**Planning & research**	**40d**	05 Oct 26	27 Nov 26
1.3	15 stakeholder problem interviews	25d	19 Oct 26	20 Nov 26
02	**Architecture & UI/UX design**	**40d**	19 Oct 26	11 Dec 26
2.3	UI/UX wireframes and screen designs	30d	26 Oct 26	04 Dec 26
03	**Dataset & annotation**	**95d**	19 Oct 26	26 Feb 27
3.2	Collect & anonymize 1,500+ messages	50d	16 Nov 26	22 Jan 27
3.4	Annotate 300+ gold requirements	35d	04 Jan 27	19 Feb 27
♦	**Frozen test set**	-	26 Feb 27	26 Feb 27
04	**App foundations & data ingestion**	**70d**	16 Nov 26	19 Feb 27
4.2	WhatsApp Android/iOS export parsing	34d	01 Dec 26	15 Jan 27
4.4	Pakistani PII masking & privacy tests	30d	04 Jan 27	12 Feb 27
05	**AI extraction & human review**	**40d**	04 Jan 27	26 Feb 27
5.2	Five-class message classification	25d	18 Jan 27	19 Feb 27
5.3	Atomic extraction, ambiguity, source IDs	25d	25 Jan 27	26 Feb 27
5.4	PM Review Queue: approve / edit / reject	25d	25 Jan 27	26 Feb 27
06	**Requirement model & baseline**	**50d**	15 Feb 27	23 Apr 27
6.2	bge-m3 embeddings + pgvector retrieval	25d	22 Feb 27	26 Mar 27
6.4	Immutable baseline lock & concurrency	25d	22 Mar 27	23 Apr 27
07	**Scope control, audit & sign-off**	**40d**	05 Apr 27	28 May 27
7.1	RAG scope classifier + cited requirements	25d	05 Apr 27	07 May 27
7.3	SHA-256 audit chain, verify, evidence pack	30d	05 Apr 27	14 May 27
7.4	Email-based client sign-off & tokens	20d	03 May 27	28 May 27
♦	**Scope + audit complete**	-	28 May 27	28 May 27
08	**Jira, Gmail & handover**	**40d**	31 May 27	23 Jul 27
8.1	Jira read-only sync & import	15d	31 May 27	18 Jun 27
8.2	Top-3 trace links, orphans, change flags	20d	14 Jun 27	09 Jul 27
8.4	Change-request drafts & handover PDF	20d	21 Jun 27	16 Jul 27
09	**Evaluation, usability & live pilot**	**60d**	07 Jun 27	27 Aug 27
9.1	Multilingual F1, hallucination, PII tests	25d	07 Jun 27	09 Jul 27
9.4	Live pilot: 2–3 agencies, ≥3 projects	20d	19 Jul 27	13 Aug 27
9.5	SUS ≥8 PMs & developer handover study	20d	26 Jul 27	20 Aug 27
♦	**Four-week pilot assessed**	-	27 Aug 27	27 Aug 27
10	**Final QA, report & defense**	**40d**	02 Aug 27	24 Sep 27
10.2	FYP report, figures & evaluation results	25d	16 Aug 27	17 Sep 27
10.3	Demo, supervisor review & final defense	10d	13 Sep 27	24 Sep 27
♦	**FYP submission**	-	24 Sep 27	24 Sep 27

1.8.Report Outline

This report describes the complete development process of ReqTrace AI. Chapter 1 presents the project introduction, background, objectives, existing solutions, and project planning. Chapter 2 covers system requirements and analysis. Chapter 3 explains system architecture, database design, and UI/UX design. Chapter 4 discusses system implementation and AI integration. Chapter 5 presents testing, evaluation, and results. Chapter 6 concludes the report with project limitations and future enhancements.

1.9.Empathy Map

The empathy map identifies the needs, concerns, and expectations of the primary user of ReqTrace AI, the Project Manager. It helps understand the challenges associated with managing client requirements and project scope.

Table 4 *Empathy Map of the Project Manager*

Category	**User Perspective**
Says	"Client requirements should be properly documented and approved."
Thinks	"How can I prevent scope creep and track requirement changes?"
Does	Reviews client messages, manages requirements, and monitors development progress.
Feels	Concerned about unclear requirements, missed changes, and project delays.
Pains	Scattered conversations, manual documentation, and missing approval records.
Gains	Automated extraction, improved traceability, faster reviews, and better scope control.

The empathy map represents the initial understanding of project managers' needs. These assumptions will be validated through stakeholder interviews. ReqTrace AI aims to address these challenges by providing AI-assisted requirement extraction, centralized documentation, and reliable scope-change tracking.